

1. From the following, which one is not among the challenges that are faced by requirement engineers during requirement elicitation?
 - a. Conflicting requirements
 - b. Obsolete requirement
 - c. requirement change
 - d. Negative stakeholders
2. One of the following statements addresses the characteristics of functional organization structure?
 - a. Limited project manager authority, limited resource availability, and a part-time project manager's role
 - b. High to almost total project manager authority, high to almost total resource availability, and fulltime project management administrative staff
 - c. Little or no project manager authority, little or no resource availability, and the functional manager control the project budget
 - d. Low to moderate project manager authority, low to moderate resource availability, and a full-time project manager's role
3. Which one is not the criteria for selecting best programming language for your problem:
 - a. Code efficiency
 - b. Modularity
 - c. Platform dependency
 - d. Portability
4. As a software tester, which testing principle can save time and cost of testing?
 - A. testing the entire development process affect time and cost
 - B. Testing early development process saves time and cost
 - c. Exhaustive testing is not possible
 - d. Prioritize testing based on risk and criticality
5. Which of the following statements designates part of the process of how a switch resolves to forward a frame destined for a recognized unicast MAC address?
 - a. It matches the unicast destination address to the bridging, or MAC address, table.
 - b. It matches the destination IP address to the destination MAC address.

- c. It matches the frame's incoming interface to the source MAC entry in the MAC address table.
- d. It matches the unicast source address to the bridging, or MAC address, table.

6. A software work product which is associated with the test execution process group is known as:

- a. Test Cases
- b. Code
- c. Requirements
- d. **Test Reports**

7. Which of the following statements is false when a packet is being compared to an-access list?

- a. There is an implicit "deny" at the end of each access list.
- b. Until all lines have been analyzed, the comparison is not over.
- c. It's always compared with each line of the access list in sequential order.
- d. **Once the packet matches the condition on a line of the access list, the packet is**

8. Which one is different from others?

- a. Requirements analysis
- b. **Requirements status tracking**
- c. Requirements validation
- d. Requirement elicitation

9. The incredible change in business and society as emerging economies develop and new technologies become available mainly:

- a. Leads to Privacy and security breaches
- b. Makes obsolete the already existing software systems
- c. Requires new testing on the existing software systems
- d. **Leads to demand for new software architecture**

10. Which of the following is not true about binary tree?

- a. **A tree is called binary tree if and only if each node has zero child.**
- b. A tree is called binary tree if each node has zero child, one child or two children
- c. Empty tree is also a valid binary tree
- d. We can visualize a binary tree as consisting of a root and two disjoint binary trees

11. A process scheduling approach on which decision as to which available process will be executed by the Processor refers to:

- ☐ a. 1/0 scheduling
- ☒ b. Short-term scheduling
- ☐ c. Long-term scheduling
- ☐ d. Medium-term scheduling

12. A modification of the free-list approach is to store the addresses of n free blocks in the first free block. ☐ a. Counting ☐ b. Linked-list ☐ c. Grouping ☐ d. Deadlock

13. Action attribute in HTML forms specifies

- ☒ a. Where to submit form
- ☐ b. The auto completion of form
- ☐ c. Which action is going on
- ☐ d. Which HTTP method is used

14. Which of the following UML diagram is not used to represent dynamic behavior of an object during analysis?

- ☐ a. Activity diagram
- ☐ b. Sequence diagram
- ☒ c. Instance diagram
- ☐ d. Collaboration diagram

15. Which of the following is a unique IPV4 address?

- ☐ a. 172.16.30.0/16
- ☒ b. 172.17.17.17/26
- ☐ c. 172.16.25.250/16
- ☐ d. 172.43.17.1/16

16. Which file system can be used to change certain kernel parameters at runtime using sysctl command?

☐ a. **Procfs** ☐ b. Ext4

☐ c. Ext3 ☐ d. Sysfs

17. Assume that you are assigned to design a dormitory management database. A dormitory is located on some building and has attributes specified as DORM (DormID, floornumber) and DormID is a primary key. Based on university's rule, six students can be assigned to one dorm. Students can be tracked with attributes STUDENT (IdNo, Name, Department) and IdNo is primary key.

Referring to this scenario, if there is a rule that says every DORM must have at least one student, which of the following is true?

☒ a. **STUDENT has existence dependency**

☐ b. STUDENT has a total participation

☐ c. DORM has partial participation

18. in a quad-tree each non-leaf node has exactly

☐ a. 1

☐ b. 3

☒ c. **4**

☐ d. 2

19. In a relation design, primary key is a critical constraint that should be considered. It should be defined :

☐ a. Non editable

☐ b. NULL ☒ c. **NOT NULL** ☐ d. Within limited value

20. A program that initializes all aspects of the system, from CPU registers to device controllers and the contents of main memory, and then starts the operating system known as:

☐ a. Bootloader ☐ b. BIOS ☒ c. **Bootstrap** ☐ d. Cache memory

21. A technique for generating plans with conditionals and loops are almost identical to those for generating programs from logical specifications is called_

☐ a. Automatic learning

☒ b. **Automatic programming**

☐ c. Automatic monitoring

- d. Automatic recursive

22. Fundamental program structure consists of the following core components except

- ☒ a. back end component
- ☐ b. Input/output features
- ☐ c. Variable definition
- ☐ d. main method

23. An input stream class in java that is used to read data from file is

- ☐ a. RandomAccessFile
- ☒ b. FileInputStream
- ☐ c. PipedInputStream
- ☐ d. FileOutputStream

24. Database approach supports different user groups who might have different interest in different parts of the database. This characteristic of data base describes:

- ☒ a. Multiple views
- ☐ b. Parallel transaction
- ☐ c. Multiple users
- ☐ d. Concurrent transaction

25. What will happen if there is no break statement next to case labels in switch statement?

- ☐ a. The program will halt
- ☐ b. The program generates exception
- ☒ c. All non-matching case labels will be executed
- ☐ d. The default block will be executed

26. Which of the following describes the leading agile development method?

- ☐ a. Extreme programming
- ☒ b. Scrum
- ☐ c. Sprint
- ☐ d. Six sigma

27. Which of the following indicates the assurance that a product, service, or result meets the needs of the customer and other identified stakeholders?

- ☐ a. SWOT Analysis
- ☒ b. Validation
- ☐ c. Verification
- ☐ d. Variance

28. Which of the following describes the functionalities of application layer?

- ☒ a. It is process to process interaction
- ☐ b. It deals mechanical, electrical and physical connectivity
- ☐ c. It deals encapsulation of frames
- ☐ d. It deals encapsulation of packets

29. Select the one that is not correct related to generalization in system modeling.

- ☐ a. common information will be maintained in one place
- ☐ b. In generalization, the attributes and operations associated with higher-level classes are also associated with the lower-level classes
- ☐ c. facilitate easy modification of data
- ☒ d. The higher-level classes are more specific than the lower-level classes by adding specific attributes and operations

30. One of the following is model optimization technique:

- ☐ a. Bagging
- ☒ b. Lasso
- ☐ c. Gradient boosting
- ☐ d. Gradient descent

31. One of the following is not purpose of software testing?

- ☒ a. Requesting more design and implementation time
- ☐ b. Improving product acceptance
- ☐ c. Enhancing reliability
- ☐ d. Identifying shortcomings

32. What of the following describes informed search algorithms?

- ☐ a. They are complete
- ☐ b. They are consistent

O c. They are admissible

O d. They are optimal

33. What will be the output of the following

program class increment { public static void

main(String args)

{

int a=4; ystem.out.print(+

+a * 5);

O a 24

O b. 5

O c. 20

O d. 25

34 A TCP/IP port number used by SSH is?

O a. 20

O b. 23

O c. 22

O d. 21

35. Select the correct statement that describes local search algorithms (LSA):

a. LSA are typically used for solving convex optimization problems.

b. The time complexity of LSA is independent of the size of the problem.

c. The quality of the solution found by LSA depends heavily on the starting point and the choice of the neighborhood function.

d. LSA always guarantee to find the globally optimal solution.

36. Which of the following is against the professional ethics of software engineering?

- a. Responsibly remain up-to-date in your profession
- b. Accepting any works regardless of your competence
- c. Respect the confidentiality of your employers or clients
- d. Use your technical skills responsibly

37. A technique that allows large numbers of ideas to be classified into groups for review and analysis.

- a. Activity-on-Node
- b. Activity List
- c. Adaptive Life Cycle
- d. Affinity Diagrams

38. Which of the following is one of the indirect applications of queues?

- a. Auxiliary data structure for algorithms
- b. Simulation of real-world queues such as lines at a ticket counter or any other first-come first-served scenario requires a queue.
- c. Operating systems schedule jobs (with equal priority) in the order of arrival (e.g., a print queue).
- d. Multiprogramming

39 Which activity is not related to data preprocessing?

- ☐ a. Data cleaning
- ☐ b. Data reduction
- ☒ c. Data optimization
- ☐ d. Data transformation

40. To reduce the dispute that might arise between the customer and contractor, each requirement should be documented and later checked during product delivery whether each documented requirements are fully implemented or not. What type of requirement validation is this?

- ☐ a. Realism check
- ☒ b. Completeness check
- ☐ c. Verifiability
- ☐ d. Validity check

41. Select the one different from others

- a. Requirement validation
- b. Requirement analysis
- c. Requirement elicitation

d. System design

42. The ultimate purpose of inheritance in OOP is:

- O a. Providing multiple forms for methods
- O b. Resources saving
- O c. Data protection

O d. Code reusability

43. Which of the following specifies divergence between Dijkstra's Algorithm (DA) and Uniform Cost Search (UCS) algorithms?

- a. UCS finds the optimal solution while DA
- b. DA discovers nodes as they come while UCS first collects them in a Queue
- c. DA is optimal, but not UCS
- d. DA first collects them into a Queue but UCS discover them as they come

44. You fit a model too closely to the particularities of the training set and obtain a model that works well on the training set but works poorly on a new data. This scenario is known as:

- O a. over fitting
- O b. under fitting
- O c. under performing
- O d. sweet spot

45. Which of the following is true regarding integration testing?

- O a. It is only performed after system testing is completed.
- O b. It uses test data that simulates ideal scenarios.
- O c. It proves the functioning of different software components when integrated together.
- O d. It emphasizes on testing individual software components in isolation.

46. Which of the following is true regarding the referential triggered action ON UPDATE CASCADE

☐ a. sets the referencing primary key of the base table to default value

☒ b. change the value of the referencing foreign key attribute to the updated primary key value for all the referencing tuples

☐ c. deletes all tuples of the referencing table

☐ d. sets the referencing primary key to NULL

47. Which object-oriented principle allows the creation of several methods with the same name that differ from each other by the type of input and output?

☐ a. Encapsulation ☒ b. Polymorphism ☐ c. Inheritance ☐ d. Overriding

48. When running your program, you did not see any error. However, the final output you got is not correct. What type of error is this?

☒ a. Syntax error

☐ b. Compiling error

☒ c. Logical error

☐ d. Run time error

49. The correct structure of the for loop statement is:

☐ a. for(increment/decrement; initialization; condition)

☐ b. for(initialization, condition, increment/decrement

☒ c. for(initialization; condition; increment/decrement)

☐ d. for(initialization; condition)

50. Which access modifier is suitable if you want your instance variables to be accessible from both the methods of the class and the descendant classes?

☐ a. Inherited ☒ b. Protected ☐ c. Public ☐ d. Private

51. A data link sub layer which is responsible for identifying Network layer protocols and then encapsulating them is known as:

☐ a. Frame relay

☐ b. Media Access Control

☒ c. Logical Link control

O d. Point to point Encapsulation

52. Which of the following keywords is used to define a variable in Javascript?

O a. const

O b. val

O c. let

O d. int

53. In software testing, a term which relates the genuine imperfect segments to the absolute number of segments anticipated as faulty known as:

a. Precision

b. Recall

c. F-measure

d. Accuracy

54 An Artificial Intelligence system is composed of :

O a. Agent and Environment

O b. Technology and Evolution O c.

Data and Information

O d. Device and Network

55. The CSS property used to specify the transparency of an element is?

O a. Background

O b. Alpha

O c. Opacity

O d. Transparency

56. If performance is the critical requirement of your architecture, which architectural approach you apply while architecting your system?

a. Increase number of components

b. Distribute operations across network

c. Increase number of component communication

d. Localize critical operations

57. Identify the one that does not affect architectural decision

- O a. The way the data is stored
- O b. Architectural style to be used
- O c. Type of the application to be developed
- O d. Expected performance of the system

58. Which of the following is not a data link layer technology?

- a. OSPF
- b. PPP
- c. Frame-relay
- d. VLAN

59. APK stands for:

- a. Android Phone Kit
- b. Android Platform Kit
- c. Android Package Kit
- d. Android Page Kit

60. Which one is not among the layouts available in android?

- O a. Relative Layout
- O b. Frame Layout
- O c. Card Layout
- O d. Linear Layout

61. Which function is used to serialize an object into a JSON string in Javascript?

- O a. stringify()
- O b. parse()
- O c. parse()
- O d. toJson()

62 The keyword used in the SELECT clause to eliminate duplicate tuples from the result of an SQL query is

- a. Group By
- b. UNIQUE
- c. PRIMARY
- d. DISTINCT

63. What does velocity in big data refer to?

- a. Speed of storage consumption
- b. Speed of data ingestion
- c. Speed of input data generation
- d. Speed of data processing and storing

64. Which one of the following is not type of Ensemble learning?

O a.

Bagging O

b. Lasso

O c.

Random forest

O d.

Boosting

65. For the software architecture to be easily understood by other design experts for later maintenance and scalability, there should be

- a. Architectural refactoring
- b. Architectural optimization techniques
- c. Architectural reverse engineering
- d. Architectural notation and semantic description

66. You have received an e-mail which is unsolicited and commercially known as?

- a. Adware
- b. Trash
- c. Spam
- d. Wana Cry

67. What is the time complexity of below function?

```
Public void function (int n) { for (int i = 0; i < n; i++)  
    for (int j = i; j < i * i; j++)  
        for (int k = 0; k < j; k++)  
            printf(" * ");
```

- a. $O(n^5)$
- b. $O(n^2)$
- c. $O(n \log n)$
- d. $O(n^7)$

68. A tool that is used to check the quality of software to ensure that it meets some minimum coding standards.

- a. Interactive Debuggers
- b. Test Data Generator
- c. Documenters
- d. Code Auditor

69. Which of the following is the first callback method that is invoked by the system during an activity lifecycle?

- ☐ a. onStart() method
- ☐ b. onClick() method
- ☒ c. onCreate() method
- ☐ d. onRestart() method

70. Security estimation is an essential component of....process

- a. SRS design
- b. Software design
- c. Risk management

d. Software development

71. Which of the following refers to unauthorized disclosure of information?

a. Authorization

b. Authentication

c. Integrity

d. Confidentiality

Confidentiality

72. Refer to the following schemas. Attributes underlined are primary keys.

HOTEL (Hotel_id integer, Hotel Name: string, Sub_city: string) ROOM (Room_id: integer, Hotel_id: integer, Price: integer)

BOOKING (Hotel_id: integer, Room_id: integer, Date from: date, Date to: date)

GUEST (Guest_id: integer, Name: string, Country: string)

Considering the above schema, which of the following SQL statement retrieves Hotel names with their respective Room_id

a.

SELECT Hotel Name, Room_Id FROM HOTEL, ROOM WHERE ROOM. Hotel_id=HOTEL. Hotel_id

b. SELECT Hotel Name, Room_Id FROM HOTEL, ROOM

c. SELECT Hotel_Name, Room_Id FROM HOTEL, ROOM WHERE ROOM. Room_id=HOTEL. Hotel_id

d. SELECT Hotel_Name, Room_Id FROM HOTEL, ROOM WHERE ROOM. Room_id=HOTEL. Hotel_id and ROOM. Hotel_id=HOTEL. Hotel_id

73. Which of the following mostly used by software project quality evaluators to using the approach of plan, check and act?

a. ISO model

ISO model

O b. Six

sigma

O c.

IEEE

O d. Capability maturity model

74. Assume you have the following hash table, implemented using linear probing. The hash function you are using is the identity function, $f(x) = x$. In which order could the elements have been added to the hash table? There are several correct answers, and you should give all of them. Assume that the hash table has never been resized, and no elements have been deleted yet.

O a. 12,9, 18, 3, 14, 21;4

• b. 9,14,4, 18, 12,3,21

O c. 12, 14, 3, 9, 4, 18, 21

O d. 12,3,14, 18,4,9,21

75. Method overriding is a characteristic of which OOP principle?

O a. Abstraction O b. Encapsulation

• C.

Inheritance

O d. Polymorphism

76. A temporary endeavor undertaken to create a unique product, service, or result is known as:

O a. Process

O b. Project

O c. Program

O d. Portfolio

77. A system software that resides in main memory is

a.Executor b. Compiler c. Loader d. Linker